A configurable generative transcriptomics framework identifies tissue-dependent synthetic utility in mouse spaceflight RNA-seq

Benchmarking conditional WGAN and diffusion models across NASA OSDR mouse transcriptomes

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Research manuscript draft for author review. Author list, acknowledgments, repository release URL, and archival DOI require final review before submission.

Abstract

Background: Mouse spaceflight studies provide access to tissues that cannot be sampled extensively from astronauts, but individual experiments are small and differ in design. We developed a configurable generative transcriptomics framework and asked whether synthetic expression could improve organ-specific flight analysis without treating generated profiles as new animals.

Methods: We assembled 1,610 mouse flight and ground-control bulk RNA-seq profiles through the NASA Open Science Data Repository API and audited all 997,515 profiles in ARCHS4 mouse. The framework varied expression transformations, feature spaces, harmonization, study scope, tissue structure, conditioning, and training regime. Paper-based WGAN-GP and diffusion implementations were screened with grouped validation, independent fidelity and effect-recovery gates, and locked testing; GeneJEPA was evaluated as a representation model because it has no expression decoder. The selected diffusion model was pretrained on 17,244 tissue-diverse ARCHS4 profiles. Downstream analysis compared pooled and tissue-specific uses of generated expression. Flight associations were tested with real profiles using accession-level random-effects models and Benjamini-Hochberg false-discovery control.

Results: None of nine matched liver harmonization arms passed all fidelity and conditional-effect gates. A calibrated study-conditioned WGAN-GP achieved correlation 0.976, precision 0.976, recall 0.994, and F1 0.985 on validation, but remained distinguishable from real data (adversarial accuracy 0.636) and failed accession-aware effect recovery; its locked test was not opened. The adapted diffusion model was the only generator to pass the final joint locked gates. Across four seeds, correlation was 0.974, precision 0.997, recall 0.996, F1 0.997, adversarial accuracy 0.475, and the Frechet-distance ratio was 0.074. Pooled augmentation reduced balanced accuracy from 0.754 with real-only training to 0.737 with real-plus-generated training. Tissue-specific use was more informative: 49 real-data BH-FDR associations also entered stable synthetic-informed selection, including a flight-lower mitotic program in thymus and a soleus program with lower *Bdh1*, *Ech1*, *Bnip3*, and *Decr1* and higher *Tpm1*.

Conclusions: Model choice depended on joint fidelity and biological-effect recovery rather than one favorable metric. Diffusion provided the strongest distributionally validated generator, but its downstream value depended on tissue and mode of use. Synthetic expression worked better as a feature prior or regularizer than as additional biological sample size. The thymus, soleus, kidney, spleen, skin, and pooled skeletal-muscle results define testable hypotheses that require independent biological replication.

Keywords: spaceflight; bulk RNA-seq; synthetic data; diffusion model; skeletal muscle; thymus; NASA OSDR; ARCHS4

Introduction

Spaceflight affects immune, musculoskeletal, metabolic, and barrier tissues through a combination of microgravity, radiation, confinement, altered nutrition, stress, and mission-specific procedures. Mouse flight experiments provide tissue access that is unavailable in astronauts, but their transcriptomic interpretation is difficult. Individual studies are small, missions differ in strain and duration, and condition labels can be entangled with study, material, genotype, or collection protocol. Pooling samples without preserving these design variables can convert study effects into apparent flight biology.

The NASA Open Science Data Repository (OSDR) now exposes sample metadata and processed assay data through a queryable biological API [1]. This makes it possible to assemble a cross-study cohort without relying on a precombined raw HDF5 object and to retain accession-level provenance. Public reference resources offer a second opportunity. ARCHS4 uniformly processes a large fraction of public human and mouse RNA-seq data [2], providing tissue-diverse reference profiles for pretraining models that would be underdetermined on OSDR alone.

Deep generative models can learn high-dimensional expression distributions. Conditional WGAN-GP models have reproduced tissue and cancer properties in GTEx and TCGA [3]. More recently, Lacan and colleagues adapted denoising diffusion probabilistic and implicit models to bulk transcriptomics and reported strong gene-correlation, neighborhood, adversarial, and downstream classification metrics [4]. GeneJEPA instead learns masked-gene representations without reconstructing expression [5]. These approaches solve different problems: a generator can sample expression, whereas a representation learner needs an additional decoder or generative objective before it can do so.

The model is only one part of the problem. Multi-study bulk RNA-seq can be represented as counts, CPM, TPM, or transformed and scaled expression; studies can be corrected, explicitly conditioned, modeled separately, or pooled. Published spaceflight workflows have used within-study standardization [24] and compared ComBat, ComBat-seq, and MBatch correction families [25]. MOBER offers a learned, inductive alternative based on an adversarial conditional variational autoencoder [26]. Any of these choices can improve one diagnostic while erasing flight-related structure or preserving study artifacts instead of biology.

We built a common framework around three model families and the preprocessing, harmonization, cohort, conditioning, and training choices surrounding them. The search was staged rather than exhaustive: inexpensive screens removed unsuitable branches before paper-duration training and locked evaluation. Models were required to pass correlation, neighborhood, adversarial, distributional, diversity, memorization, and FLT/GC-effect gates independently. No composite score allowed one strong metric to compensate for another weak one.

Synthetic expression is commonly presented as a remedy for small sample size. Generated profiles, however, are not new biological replicates. After model selection, we separated three downstream questions: whether one pooled augmentation strategy helped at all, whether different tissues benefited from different synthetic-data uses, and whether prioritized genes were associated with flight in real samples.

Our primary biological questions were whether this approach could clarify the tissue response to spaceflight and whether anatomical separation would expose muscle-specific responses hidden by pooling.

The tissue-specific analysis supplied its most coherent biological programs in thymus and soleus. Kidney, spleen, skin, pooled skeletal muscle, and adrenal gland supplied narrower candidates. Results from the remaining tissues define the exploratory boundary of the approach.

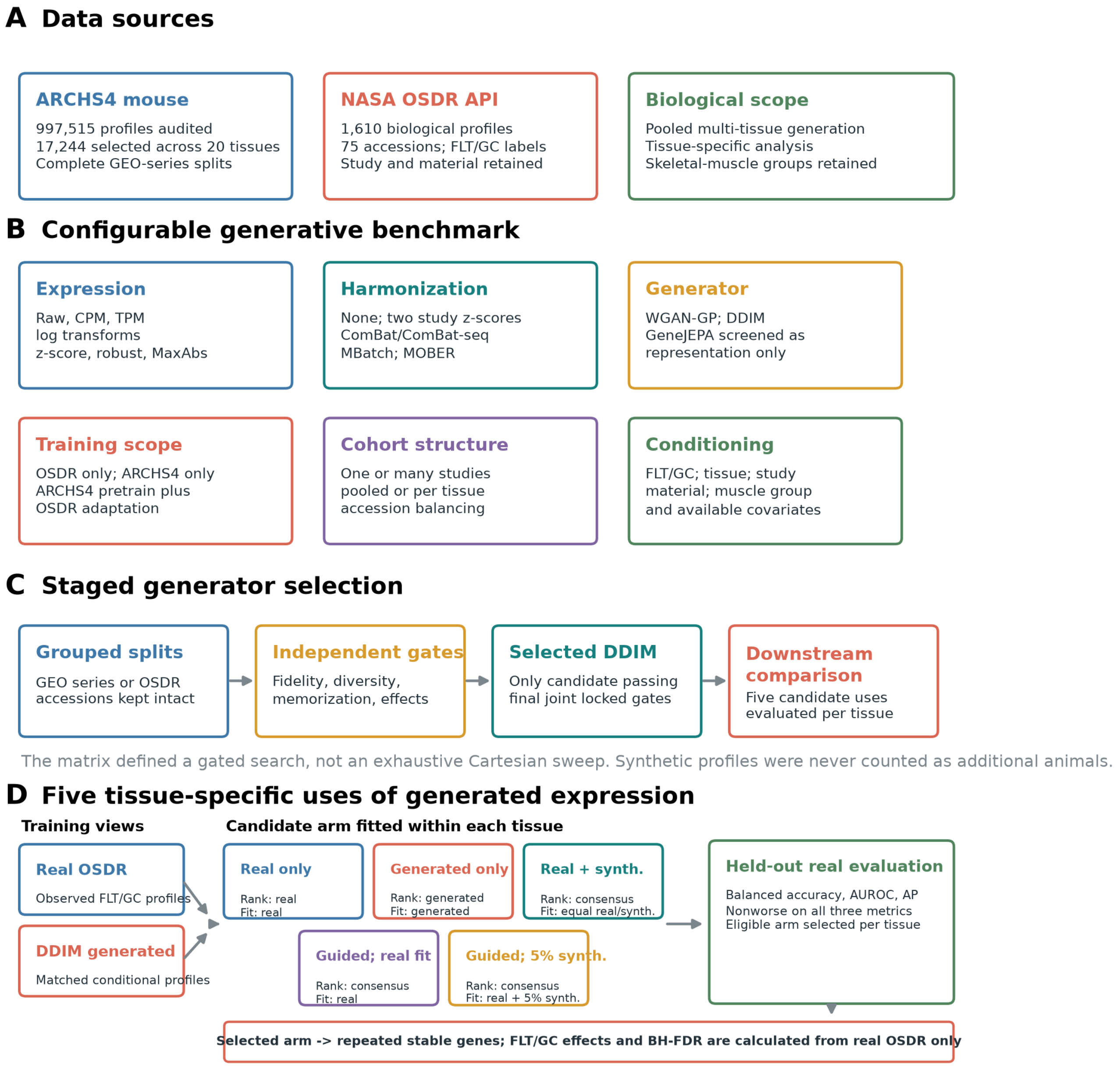

Figure 1. Configurable generative transcriptomics framework. (A) ARCHS4 supplied tissue-diverse reference profiles and the NASA OSDR API supplied flight and ground-control profiles with study provenance. (B) Configurable axes covered expression representation, harmonization, model family, training scope, cohort structure, and conditioning. (C) GEO series and OSDR accessions were grouped during generator selection. (D) Each tissue compared real-only, generated-only, real-plus-generated, consensus-ranked real-only, and consensus-ranked low-weight synthetic training. Every arm was evaluated on held-out real profiles; biological effects and false-discovery rates came from real OSDR samples.

Materials and methods

Data sources

The OSDR Biological Data API was used to identify *Mus musculus* bulk RNA-seq assays with flight or ground-control labels [1]. Tissue and material names were harmonized while preserving study provenance. The resulting cohort contained 1,610 biological profiles from 75 accessions: 835 flight and 775 ground control. Full-transcriptome expression was converted to transcripts per million before selecting a 974-gene mouse landmark panel. No raw integrated OSDR H5 file was used.

The local ARCHS4 mouse resource contained 997,515 public RNA-seq profiles [2]. All nine GEO series linked from the API-derived OSDR metadata were excluded before reference selection, removing 108 otherwise eligible profiles and preventing cross-resource sample overlap. A healthy-preferred, tissue-balanced subset of 17,244 profiles spanning 20 tissue classes was then used for model pretraining. Complete GEO series were assigned to one reference role, producing 10,150 training, 2,466 validation, and 4,628 test profiles with no series overlap. This backbone was used for both the held-out test and the broad development screens.

Table 1. Data scope.

Source	Profiles used	Biological scope	Role
ARCHS4 mouse v2.5	17,244	20 tissue classes	Mouse tissue pretraining
NASA OSDR API	1,610	75 accessions; 835 flight and 775 ground control	Spaceflight analysis

Configurable generative benchmark

The framework separated decisions that are often bundled into one model configuration. Expression could be represented as raw counts, CPM, or TPM and then left untransformed or subjected to log, z-score, robust, or MaxAbs transformations. Feature spaces included all shared genes, fold-selected highly variable genes, Reactome genes, and mapped mouse L1000 landmarks. Cohorts could contain one study, a selected multi-study subset, or every eligible accession. Models could be pooled with tissue conditioning or fitted per tissue, with optional FLT/GC, study, material, muscle-group, sex, assay, platform, and source inputs.

Three training regimes were available: OSDR only, ARCHS4 only, and ARCHS4 pretraining followed by OSDR adaptation. The design matrix also included unconditional controls and study-conditioned versus study-unconditioned models. It defined a staged search rather than an exhaustive Cartesian sweep. Smoke tests and lower-cost screens removed branches before paper-duration training, repeat evaluation, and locked testing.

Table 2. Configurable pipeline and selected analysis branch.

Axis	Alternatives represented in the framework	Selected branch for downstream analysis
Expression	Raw, CPM, TPM; log1p/log2p1; z-score, robust, or MaxAbs scaling	Full-transcriptome TPM, then train-fitted MaxAbs scaling
Feature space	All shared genes, fold-selected HVGs, Reactome genes, mapped mouse L1000	974 mapped mouse L1000 landmarks
Harmonization	None, two study-wise z-score schemes, ComBat, ComBat-seq, three MBatch methods, MOBER	No global batch correction; accession represented as a condition
Training data	OSDR only, ARCHS4 only, ARCHS4 pretraining plus OSDR adaptation	ARCHS4 pretraining plus OSDR adaptation
Cohort structure	One or multiple studies; pooled or per-tissue fitting	All eligible OSDR accessions in a pooled tissue-conditioned generator
Conditions	FLT/GC, tissue, study, material, muscle group, and available design covariates	Tissue, FLT/GC, accession, and material
Model	WGAN-GP, DDIM, GeneJEPA representation screen	Factorized conditional DDIM
Validation	Grouped validation, repeat generation, locked test, unconditional controls	Four-seed 293-profile locked OSDR test

Preprocessing and harmonization

Each generator received its paper-based preprocessing and a shared benchmark representation. The WGAN-GP branch used log-transformed expression with training-gene z-scores [3]. The diffusion branch used TPM, a mapped mouse landmark panel, and training-fitted MaxAbs scaling [4]. GeneJEPA used its sparse log-transformed token representation and a global nonzero-expression standardization [5]. All feature selection and transform statistics were fitted on training partitions.

Nine harmonization arms were compared in a matched liver diffusion experiment: no correction, the within-study standardization used by Ilangovan et al. [24], a mentor-proposed within-study then global z-score, ComBat, ComBat-seq, MBatch Median Polish, MBatch Empirical Bayes, MBatch ANOVA [25], and MOBER [26]. ComBat, ComBat-seq, and MBatch do not provide a natural frozen transform for a new batch, so their held-out applications were labeled transductive sensitivity analyses. MOBER supplied an inductive learned projection. A harmonizer could advance only if it preserved both expression fidelity and FLT/GC effects; reduced study separation alone was insufficient.

Model implementations and training regimes

The WGAN-GP reproduced the Viñas et al. generator and critic topology: 64-dimensional noise, two 256-unit hidden layers, five critic updates per generator update, gradient-penalty weight 10, and the released RMSProp settings [3]. The diffusion branch reproduced the 974-landmark architecture of Lacan et al., with two 8,192-unit hidden blocks, 1,000 diffusion timesteps, quadratic noise schedule, Adam optimization, OneCycle scheduling, mixed precision, and exponential moving average [4,12,13]. The ARCHS4 reference model was trained for 15,000 epochs on an NVIDIA A100.

GeneJEPA was evaluated with its released 768-dimensional, 24-block architecture [5]. It has no expression decoder, so it could be assessed for tissue representation but could not produce synthetic RNA-seq profiles. It was not treated as a third generator, and no unvalidated decoder was attached.

Model validation and selection

Complete GEO series were grouped in ARCHS4 splits, and complete OSDR accessions were grouped for model selection where the protocol required study-level separation. Candidate generators were evaluated with gene-correlation agreement, precision, recall, F1, external real-versus-synthetic adversarial accuracy, and Frechet distance in a real-fitted mouse expression embedding [14,15]. Diversity and nearest-neighbor memorization were tested separately. Conditional models also had to recover pooled FLT/GC effects and accession-aware skeletal-muscle effects.

The nominal fidelity gates were correlation at least 0.98, precision at least 0.95, recall at least 0.85, F1 at least 0.90, adversarial accuracy from 0.40 to 0.60, and Frechet distance no greater than the 95th percentile of real-versus-real splits. For small OSDR partitions, correlation was also compared with a prespecified finite-sample real-bootstrap floor; the absolute 0.98 target remained reported. Metrics were gated independently, with no composite score or compensating rank. A locked test was opened only after a candidate passed its preceding selection stage.

Selected conditional diffusion model

The selected generator was the only candidate to pass the final joint locked gates. It used the 17,244-profile ARCHS4 backbone and a factorized OSDR adaptation with tissue, FLT/GC, accession, and material inputs. This allowed generation of flight or ground-control expression for represented tissue and study contexts. The broad adaptation supplied the repeated all-tissue and anatomical-muscle development screens. Exact training stages, seeds, and hardware records are provided in the supplementary methods.

Evaluation funnel and synthetic-guided analysis

Evaluation used three stages. First, we tested the most direct use of synthetic expression: pooled augmentation across all tissues. Individual tissue profiles were not averaged or collapsed; instead, they were combined in one FLT/GC classification problem governed by a common decision rule. The locked benchmark compared classifiers trained with real profiles, generated profiles, or real plus generated profiles. Second, because the pooled strategy did not improve performance, each tissue was evaluated separately with five candidate uses (Fig. 1D). Real-only ranking and fitting used observed OSDR profiles; generated-only ranking and fitting used DDIM profiles. Real-plus-generated training used consensus ranking and equal total real and synthetic weight. Both guided arms used real/synthetic consensus ranking: one fitted the classifier only on real profiles, while the other added condition-recentered generated profiles at 0.05 total synthetic weight. Nested development splits withheld profiles but retained representation from the same accessions. These are within-study development results.

Synthetic attribution was retained only when the selected arm was nonworse than real-only training across balanced accuracy, AUROC, and average precision under the frozen eligibility rule.

Third, stable features from the selected tissue arm were compared with stable real-only features. Genes stable under both approaches were interpreted as reinforced. Genes stable only under the eligible synthetic-informed arm were interpreted as synthetic-promoted. "Synthetic-promoted" describes repeated feature selection; it does not mean that a gene was absent from real expression or biologically novel.

Flight-minus-ground effects were estimated within each OSDR study and then summarized with a random-effects model [16]. This kept mission-level contrasts separate before meta-analysis and prevented generated profiles from increasing the biological sample count.

Within each tissue, the 974 real-data gene-level meta-analysis P values were adjusted with the Benjamini-Hochberg procedure, and BH FDR below 0.05 defined the primary statistical inclusion rule [6]. Generated profiles were never entered as biological replicates. Synthetic selection status was recorded separately as reinforced, synthetic-promoted, real-only, or not stably selected. Accession-direction agreement, between-study heterogeneity, and leave-one-accession-out results were retained as interpretation and sensitivity measures rather than inclusion requirements. The complete BH-FDR inventory is provided in Supplementary Table S11, its synthetic-guided subset in Table S10, and tissue-level counts in Table S12. Reactome was used to group selected genes into biological processes [7].

Interpretation of the three stages

The stages answer different questions and should not be collapsed into one performance claim. The pooled benchmark tests a common augmentation strategy. Tissue screens identify where and how synthetic data may be useful. Real-only random-effects tests establish association in the observed studies.

Table 3. Evaluation stages and permitted interpretation.

Stage	Analysis	Data separation	Question answered	Permitted interpretation
1	Pooled utility benchmark	Locked profiles from represented studies	Does one synthetic-data policy help across tissues?	Method-level positive or negative result; no biological claim
2	Tissue-specific five-arm screen	Held-out profiles within represented accessions	Which synthetic use, if any, helps each tissue?	Developmental utility and stable feature nomination
3	Real-data random-effects BH FDR	FLT-GC effects estimated separately within each accession	Are nominated genes associated with flight in observed studies?	Real biological association; synthetic status reported separately

Signals supported by Stages 2 and 3 are described as developmental or exploratory because the tissue screens do not establish generalization to a new study.

Results

Broad-reference and representation screens defined the starting point

The ARCHS4 diffusion model retained broad tissue information. Classifiers trained on real and synthetic reference profiles each achieved balanced accuracy 0.781 on 4,628 held-out profiles from complete GEO series. Precision was 0.951, recall 0.890, F1 0.919, adversarial accuracy 0.515, and the Frechet-distance ratio was 0.866. Gene-correlation agreement was 0.878 and failed its prespecified floor. We therefore retained the model only as tissue-conditioned initialization and required OSDR adaptation to pass a separate validation sequence.

The GeneJEPA duration screen achieved held-out tissue balanced accuracy 0.703, compared with 0.839 from expression in that screen. This indicated that the representation contained tissue information but did not outperform the input expression. More importantly for the present objective, the released model had no expression decoder and could not enter the synthetic-generation comparison.

Harmonization improved individual metrics but not the full conditional objective

The matched liver benchmark used the same conditional diffusion architecture, seed, 974 genes, and 119/50 training/validation profiles for each arm; the 70-profile locked test was not used for this screen. None of the nine harmonization strategies passed all independent fidelity, pooled-condition, and accession-effect gates (Fig. 2D; Supplementary Tables S13-S14).

The mentor two-stage z-score was the most balanced conventional arm, with correlation 0.348, precision 0.440, recall 1.000, F1 0.611, adversarial accuracy 0.690, and Frechet ratio 0.977. MOBER produced the highest correlation, 0.808, but precision was 0.260, F1 was 0.413, adversarial accuracy was 0.770, and the Frechet ratio was 33.311. ComBat, ComBat-seq, and the MBatch methods had correlations near zero and Frechet ratios from 44.716 to 205.470. Thus better alignment on one statistic did not produce a usable conditional generator. The final path retained TPM/MaxAbs expression without global batch correction and represented accession explicitly during OSDR adaptation.

Diffusion passed the joint gates that rejected WGAN-GP

The calibrated study-conditioned WGAN-GP had strong validation correlation, precision, recall, and F1: 0.976, 0.976, 0.994, and 0.985 across six generation seeds. Its Frechet ratio was also low at 0.144. However, external adversarial accuracy was 0.636, outside the accepted 0.40-0.60 interval, so real and generated profiles remained separable. Pooled FLT/GC-effect recovery passed in all six repeats, but the skeletal-muscle accession-aware gate passed in none. No WGAN repeat passed the full fidelity gate, and its locked test remained unopened.

The OSDR-adapted DDIM passed all six locked fidelity criteria in each of four generation repeats. Mean correlation was 0.974, precision 0.997, recall 0.996, F1 0.997, adversarial accuracy 0.475, and Frechet ratio 0.074. Correlation remained below the absolute paper target of 0.98 but exceeded the prespecified finite-sample real-bootstrap floor in all four repeats. Pooled FLT/GC-effect recovery passed in three of four repeats, and skeletal-muscle accession-effect recovery passed in all four. Diffusion was therefore selected because it passed the final joint gates, not because every individual value was numerically better than WGAN-GP (Table 4).

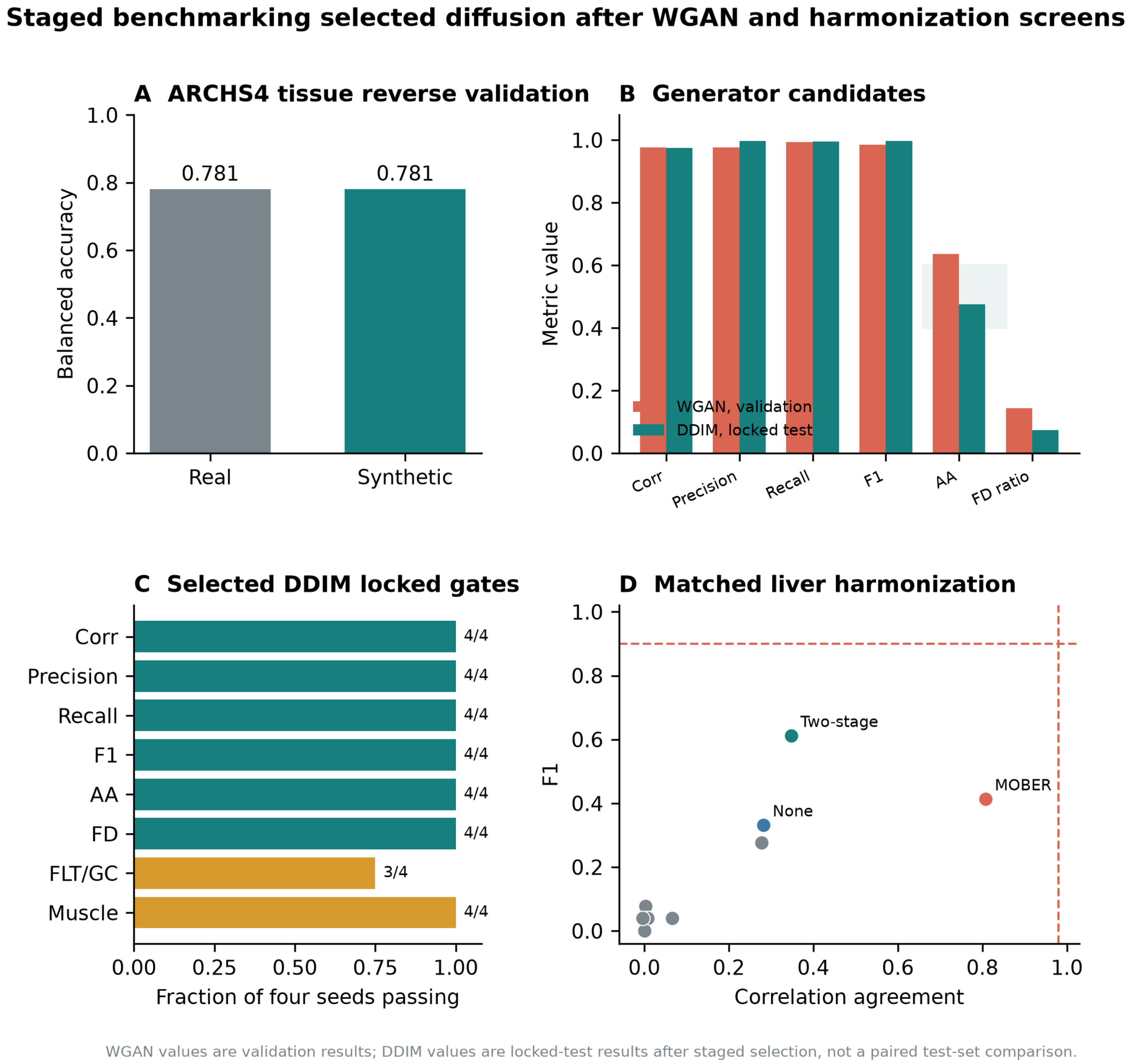

Figure 2. Generator benchmarking and selection. (A) Real-trained and synthetic-trained classifiers retained the same broad ARCHS4 tissue balanced accuracy. (B) Calibrated WGAN-GP validation metrics and DDIM locked-test metrics; these are consecutive selection stages, not a paired test-set comparison. The shaded band marks the accepted adversarial-accuracy interval. (C) Fraction of four locked DDIM generation seeds passing each fidelity and effect-recovery gate. (D) Correlation and F1 in the nine-arm matched liver harmonization benchmark. Dashed lines mark the absolute targets; no harmonization arm passed the joint criteria.

Table 4. Staged generator selection. WGAN-GP failed validation, so its locked test was not opened; DDIM values are from the later locked test and are not a paired comparison on the same split.

Model	Evaluation split	Corr.	Precision	Recall	F1	AA	FD/ real P95	FLT/ GC gate	Accession gate	Decision
Broad-reference DDIM	4,628 held-out ARCHS4 profiles	0.878	0.951	0.890	0.919	0.515	0.866	NA	NA	Retained only as initialization
Study-conditioned WGAN-GP	536-profile validation; 6 seeds	0.976	0.976	0.994	0.985	0.636	0.144	6/6	0/6	Rejected; test unopened
Factorized DDIM	293-profile locked OSDR test; 4 seeds	0.974	0.997	0.996	0.997	0.475	0.074	3/4	4/4	Selected

Diffusion generation recovered tissue-conditioned expression structure

The reverse trajectory provides a direct view of the model transforming noise into tissue-conditioned expression. In the ARCHS4 reference model, profiles moved from an overlapping noise cloud at timestep 1,000 toward the tissue-structured real-data manifold at timestep 0 (Fig. 3, top). After OSDR adaptation, locked real and generated profiles occupied similar tissue-defined regions, and flight and ground-control profiles remained interspersed within that broader structure (Fig. 3, bottom). These two-dimensional projections are descriptive: model selection used the full correlation, precision, recall, adversarial, Frechet-distance, and conditional-effect gates rather than visual similarity.

Pooled augmentation motivated tissue-specific analysis

We first asked whether synthetic profiles could simply expand one multi-tissue FLT/GC training cohort. On the locked OSDR test, balanced accuracy was 0.754 with real-only training, 0.695 with generated-only training, and 0.737 with real-plus-generated training. The corresponding AUROCs were 0.820, 0.751, and 0.791. Pooled augmentation therefore did not improve the broad classifier.

This result is consistent with strong tissue variation in bulk expression and with flight responses that differ among tissues. A common classifier can dilute tissue-specific condition signals even when the generator itself is conditioned on tissue. We therefore moved from a single pooled augmentation policy to separate tissue-level analyses. When synthetic use was selected separately by tissue, several arms improved balanced accuracy, AUROC, and average precision within represented studies (Table 5). Different tissues selected different uses: spleen, skin, and soleus selected real-plus-generated training; pooled skeletal muscle selected feature guidance with a real-only classifier; kidney and thymus selected low-weight guided training; and lung and adrenal gland selected generated-only training. These are development-screen results because profiles, rather than complete studies, were withheld.

Table 5. Selected tissue-specific development results. Every displayed synthetic-informed arm met the balanced-accuracy, AUROC, and average-precision eligibility rule. Complete canonical-tissue and muscle-group results are in Supplementary Tables S9 and S6.

Tissue	Selected synthetic use	Balanced accuracy, real to selected	AUROC, real to selected
Thymus	Low-weight guided training	0.733 to 0.844	0.818 to 0.887
Skeletal muscle, pooled	Feature-guided real-only classifier	0.861 to 0.932	0.924 to 0.961
Soleus	Real plus generated	0.925 to 0.963	0.980 to 0.980
Kidney	Low-weight guided training	0.654 to 0.707	0.680 to 0.771
Spleen	Real plus generated	0.517 to 0.648	0.540 to 0.704
Skin	Real plus generated	0.671 to 0.756	0.691 to 0.768
Lung	Generated only	0.664 to 0.742	0.680 to 0.830
Adrenal gland	Generated only	0.781 to 0.922	0.906 to 0.984

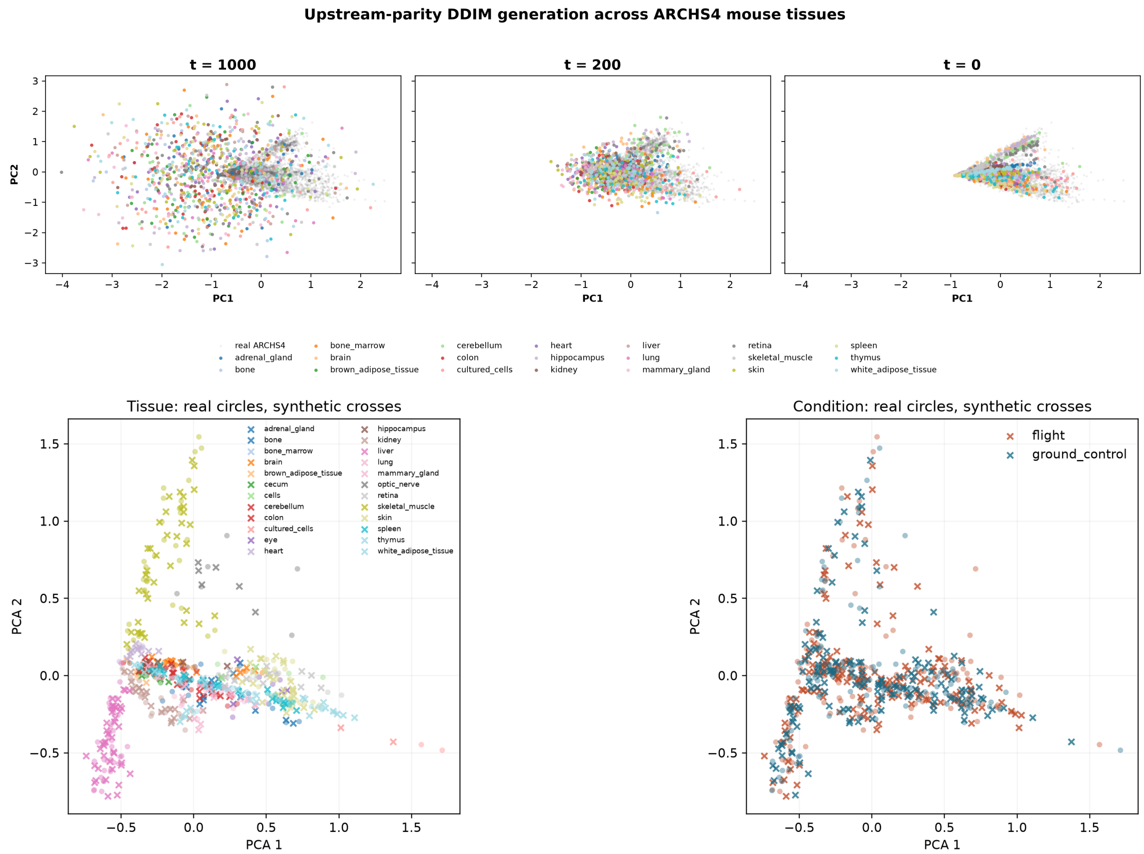

Figure 3. Diffusion generation across reference pretraining and OSDR adaptation. Top: ARCHS4 tissue-conditioned profiles at DDIM timesteps 1,000, 200, and 0 in a PCA space fitted to real reference expression; gray points are real ARCHS4 profiles and colors identify generated tissue conditions. Bottom: locked OSDR profiles for generation seed 5020; circles denote real profiles and crosses denote generated profiles, colored by tissue on the left and flight condition on the right. PCA views are descriptive and do not replace the quantitative validation gates.

Synthetic-informed selection identified real-data associations

The real-data screen yielded 202 BH-FDR tissue-gene associations across 10 of 22 canonical tissue analyses and 257 across the five anatomical muscle groups. These are tissue-gene results rather than 459 unique or independent discoveries: a gene can be significant in more than one tissue, and pooled skeletal muscle overlaps its anatomical subgroups. Forty-nine associations also entered stable synthetic-informed selection: 26 were synthetic-promoted and 23 were reinforced by both real-only and synthetic-informed selection.

All 459 P values and FDR values came from real profiles. Synthetic data affected only whether a gene was repeatedly prioritized and, for eligible augmentation arms, classifier fitting. Synthetic labels were suppressed where the selected arm failed its metric gate; quadriceps, EDL, and liver therefore contribute real-data associations but no synthetic-informed claims. The complete inventory, including associations with mixed study directions, is provided in Supplementary Tables S10-S12.

Synthetic-informed thymus analysis identifies lower proliferative renewal

The thymus screen selected low-weight synthetic-guided training. Balanced accuracy increased from 0.733 to 0.844, AUROC from 0.818 to 0.887, and average precision from 0.862 to 0.918 across repeated within-study splits. Sixteen thymus associations passed real-data BH FDR and entered stable synthetic-informed selection: 13 were synthetic-promoted and three were reinforced.

The strongest coherent subset contained flight-lower *Nusap1*, *Stmn1*, *Birc5*, *Cdk1*, *Top2a*, *Ccnb2*, *Aurka*, and *Ccne2*, all promoted by synthetic guidance, together with reinforced *Ube2c* and *Gmnn* (Fig. 4A). Each had a lower random-effects estimate across the five thymus accessions and BH FDR below 0.05. Reactome analysis of the synthetic-supported set identified mitotic cell cycle, DNA replication, S phase, APC/C regulation, and G2/M control (Fig. 4B).

These genes were present and statistically supported in the real data. Synthetic guidance changed their repeated selection behavior and organized them into a coherent panel; it did not create independent evidence or establish de novo biological discovery.

Prior thymus studies report related biology, although the present result is more specific. STS-135 mouse thymus showed changes in cell-cycle and DNA-damage programs, including lower checkpoint-related expression [8]. A later ISS experiment reported marked thymus mass loss and partial artificial-gravity rescue of cell-cycle expression [9]. The current signature emphasizes mitotic completion and replication rather than acute apoptosis alone. Agreement across accessions suggests a shared thymus response despite study heterogeneity, but it does not identify the responsible cell population.

Bulk thymus expression cannot distinguish lower transcription within proliferating thymocytes from loss or redistribution of proliferating cell populations. The defensible biological conclusion is lower abundance of a mitotic transcript program in flight, consistent with reduced proliferative renewal. Cell-resolved or histological confirmation is required to assign the effect to a cell-intrinsic mechanism.

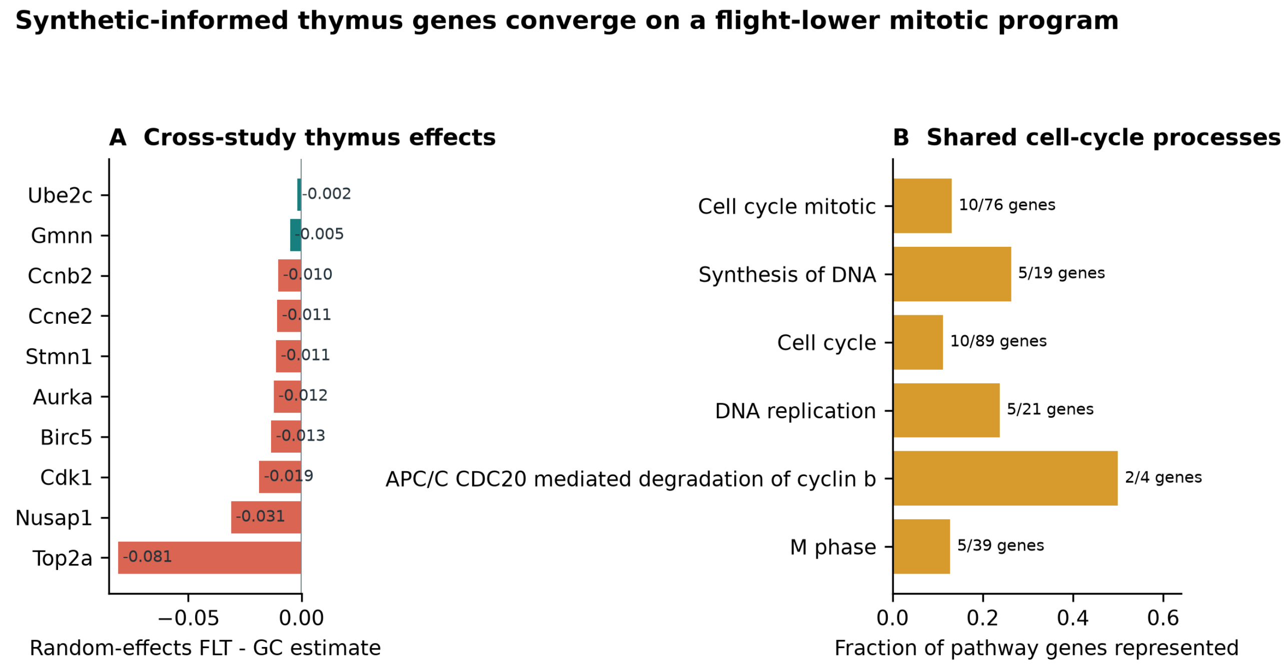

Figure 4. Thymus response to spaceflight. (A) Real-data random-effects flight-minus-ground estimates for ten genes across five thymus accessions; coral denotes synthetic-promoted genes and teal denotes reinforced genes. (B) The synthetic-supported set converges on mitotic and DNA-replication processes.

Anatomical separation exposes a soleus-specific metabolic program

Aggregate skeletal muscle concealed substantial anatomical heterogeneity. We therefore examined extensor digitorum longus, gastrocnemius, quadriceps, soleus, and tibialis anterior separately. Soleus produced the clearest biological pattern: its selected genes showed consistent flight effects across three accessions and converged on related metabolic processes.

The soleus screen selected real-plus-generated training. Balanced accuracy increased from 0.925 to 0.963, AUROC remained 0.980, and average precision increased from 0.980 to 0.986. Five BH-FDR genes were stable in both real-only and synthetic-guided selection: *Bdh1*, *Ech1*, *Bnip3*, and *Decr1* were lower in flight in all three accessions, while *Tpm1* was higher (Fig. 5B). Four genes passed leave-one-accession-out FDR; *Decr1* did not.

The model did not promote a soleus gene absent from stable real-only selection. Its contribution was reinforcement of a coherent existing pattern. Reactome connected the retained genes to mitochondrial protein turnover, mitochondrial fatty-acid beta-oxidation, and lipid metabolism (Fig. 5C). *Bdh1* and *Ech1* support oxidative substrate handling, *Bnip3* supports mitochondrial quality control, *Decr1* contributes to unsaturated fatty-acid oxidation, and *Tpm1* suggests contractile remodeling.

Prior 30-day spaceflight profiling of mouse soleus reported a slow-to-fast shift and broad changes in oxidative metabolism, PPAR signaling, and contractile genes [10]. Unloading studies have also reported reduced soleus fatty-acid oxidation [11]. This is literature-supported panel reinforcement rather than de novo gene discovery.

Unlike thymus, soleus was represented during model development. Its cross-study consistency makes it a focused biological hypothesis, but an entirely unseen soleus study is still needed for independent confirmation.

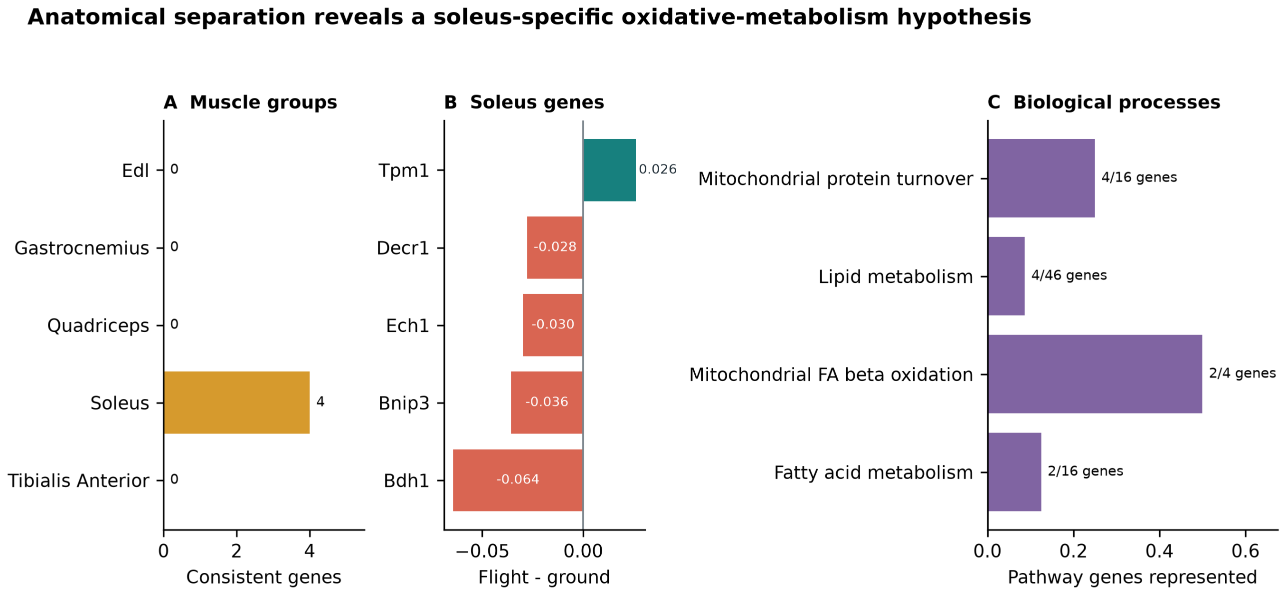

Figure 5. Skeletal-muscle and soleus response. (A) Number of synthetic-informed genes that also pass real leave-one-accession-out FDR in each anatomical muscle group. (B) Five reinforced soleus BH-FDR genes with consistent real flight effects. (C) Their strongest shared biological processes center on mitochondrial turnover and lipid metabolism.

Other muscle groups provide narrower hypotheses

The pooled skeletal-muscle screen selected synthetic-guided feature ranking and improved balanced accuracy, AUROC, and average precision by 0.071, 0.036, and 0.037. Twelve BH-FDR genes were synthetic-informed, nine of which passed leave-one-accession-out sensitivity. The associated gene sets included interferon signaling and sialic-acid metabolism. Because pooled muscle combines anatomical groups with distinct physiology, this result is interpreted alongside, rather than instead of, the soleus analysis.

Among the remaining anatomical groups, tibialis anterior retained a real-plus-generated arm and gastrocnemius retained a low-weight guided arm. Tibialis anterior reinforced *Cdkn1a*, *St3gal5*, and *Bnip3* and promoted *Cebpd*; gastrocnemius promoted *Fhl2* and *Nfkbia*. None of these genes passed leave-one-accession-out FDR. EDL and quadriceps selected real-only arms, so their BH-FDR genes remain in the complete real-data inventory but are not attributed to the synthetic workflow.

Spleen screen nominates adhesion and cytoskeletal genes

The spleen screen selected real-plus-generated training. Balanced accuracy increased from 0.517 to 0.648, AUROC from 0.540 to 0.704, and average precision from 0.589 to 0.749. Synthetic guidance promoted flight-higher *Rai14*, *Ptprk*, and *Myl9* and reinforced flight-higher *Loxl1*. *Rai14*, *Ptprk*, and *Loxl1* had the same direction in all six studies; *Myl9* agreed in five of six. None passed leave-one-accession-out FDR, and the set did not produce significant Reactome enrichment.

The four genes provide a tentative structural hypothesis rather than a pathway claim. RAI14 links actin-associated mechanosensing to Hippo signaling, PTPRK regulates cell-cell junctions, MYL9 contributes to actomyosin contractility, and LOXL1 supports elastic extracellular-matrix maintenance [21-23]. Their shared direction is compatible with altered adhesion, mechanics, or tissue architecture, but the present bulk data cannot identify a common source cell or establish one mechanism.

Other organ responses were heterogeneous

Lung produced large repeated development-screen gains and cell-cycle, senescence, and PI3K/AKT-related enrichment, but no selected gene passed the real-data BH-FDR screen. We therefore treat the lung result as predictive development evidence without a supported gene-level biological claim.

Skin selected real-plus-generated training and promoted flight-higher *Plscr1*. *Plscr1* passed BH FDR and had the same direction in all six studies, although it did not pass leave-one-accession-out FDR. Broader selected sets were enriched for G1/S and DNA-repair processes, consistent with prior OSDR skin analyses reporting strong mission, strain, recovery-interval, and anatomical-site dependence [17]. This is a literature-aligned developmental candidate rather than a confirmed transferable signal.

Kidney produced the clearest secondary gene-level result. *Slc37a4* was reinforced by both selection arms and was higher in flight in all six kidney accessions. Synthetic guidance also promoted flight-higher *Inpp4b*; it passed BH FDR and remained significant in leave-one-accession-out analysis, although one of six accession effects was negative. The selected pair did not yield significant Reactome enrichment. *Slc37a4* supports renal glucose handling, while *Inpp4b* nominates phosphoinositide signaling. Both require independent study confirmation and should be interpreted alongside prior reports of strain-dependent lipid, ECM, TGF-beta, Wnt, and nephron-remodeling responses [18-20].

Adrenal gland produced flight-lower synthetic-promoted *Psmb8* and reinforced *Tspan4*, each with the same direction in all three studies but neither passing leave-one-accession-out FDR. Retina showed repeated predictive gains and selected-set enrichment but no synthetic-informed BH-FDR gene. Liver selected the real-only arm: its 19 real-data BH-FDR genes remain in the complete inventory, but none support a synthetic-guided claim. Quadriceps and EDL likewise selected real-only arms.

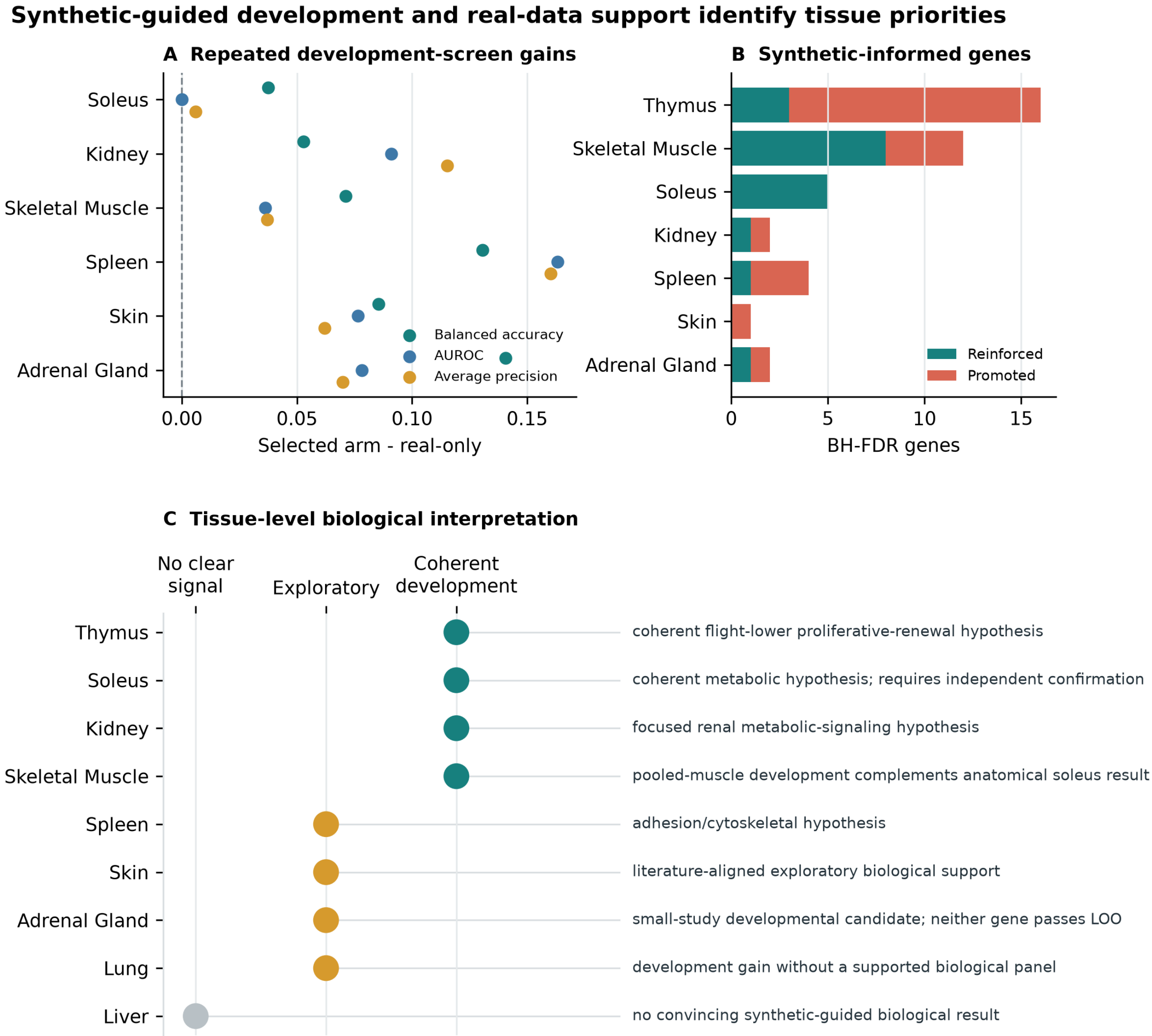

Figure 6. Tissue evidence. (A) Repeated development-screen changes relative to real-only models for selected tissues. (B) BH-FDR genes promoted or reinforced by the synthetic workflow; synthetic labels are suppressed where the generated arm failed its metric gate. (C) Thymus and soleus provide the clearest process-level hypotheses, kidney supplies a focused secondary pair, and spleen, skin, and adrenal gland remain exploratory.

Table 6. Selected synthetic-guided biological interpretations by tissue. Complete real-data BH-FDR results are in Supplementary Tables S11-S12.

Tissue	Main signal	Interpretation
Thymus	Lower mitotic genes; APC/C, G2/M, and DNA replication	Coherent synthetic-informed panel supported by real-data BH FDR
Soleus	Reinforced FLT-lower *Bdh1*, *Ech1*, *Bnip3*, and *Decr1*; FLT-higher *Tpm1*	Coherent mitochondrial and lipid-metabolism program; no promoted gene
Skeletal muscle, pooled	12 synthetic-informed BH-FDR genes; interferon and sialic-acid terms	Developmental complement to the anatomically specific soleus result
Kidney	Promoted *Inpp4b* and reinforced *Slc37a4*, both FLT-higher	Focused renal metabolic-signaling hypothesis
Spleen	Promoted *Rai14*, *Ptprk*, and *Myl9*; reinforced *Loxl1*	Developmental adhesion/cytoskeletal hypothesis
Skin	Promoted FLT-higher *Plscr1* plus cell-cycle/DNA-repair enrichment	Literature-aligned developmental candidate
Adrenal gland	Promoted *Psmb8* and reinforced *Tspan4*, both FLT-lower	Three-study developmental candidate
Tibialis anterior	Reinforced *Cdkn1a*, *St3gal5*, and *Bnip3*; promoted *Cebpd*	Exploratory stress and metabolic response
Gastrocnemius	Promoted *Fhl2* and *Nfkbia*	Exploratory two-gene result without a coherent pathway
Lung	Generated-only development improved all three predictive metrics	Predictive development result without a BH-FDR gene in the 974-gene panel
Retina	Predictive and pathway gains without a synthetic-informed BH-FDR gene	Exploratory gene/pathway mismatch
Quadriceps, EDL, liver	Screen retained real-only arms	Real-data BH-FDR associations are not synthetic-guided findings

Discussion

Generator selection required more than distributional correlation

The benchmark separated generative quality into independent requirements. The WGAN-GP reproduced mean structure and local neighborhoods well, and its correlation was slightly higher than the final DDIM mean. It nevertheless remained distinguishable by an external real-versus-synthetic classifier and did not recover skeletal-muscle effects consistently across accessions. Conversely, the broad ARCHS4 DDIM preserved tissue identity and passed most distributional tests while missing the correlation-structure gate. Neither result alone justified biological use.

OSDR adaptation changed that balance. The factorized DDIM was the only candidate to combine near-chance external adversarial accuracy, high precision and recall, low relative Frechet distance, pooled FLT/GC recovery, and accession-aware muscle-effect recovery. The WGAN and DDIM values were obtained at consecutive stages because the WGAN failed before the locked test was opened. The evidence supports the selected diffusion lineage; it does not establish that diffusion will outperform adversarial training under every matched architecture or dataset.

The harmonization benchmark reached a similar conclusion. Study-wise transformations and learned correction could improve individual diagnostics, but none preserved the complete conditional objective on the fixed liver split. MOBER's high correlation coexisted with poor neighborhood precision, external separability, and excessive distributional distance. The two-stage z-score was more balanced but still failed the fidelity and effect gates. Explicitly representing study context proved more defensible here than forcing one globally corrected expression space.

What synthetic data added

One straightforward way to use synthetic expression is pooled augmentation: add generated profiles from every tissue to one broad FLT/GC training cohort. That approach did not improve classification. Tissue identity is a major source of expression variation, and a shared classifier can obscure condition responses that use different genes or directions in different organs. We therefore treated the failed pooled experiment as motivation for tissue-specific analysis rather than as evidence that synthetic data had no downstream value. In tissue-specific development, direct augmentation, low-weight augmentation, generated-only training, and feature guidance each helped at least one tissue. No single use of synthetic expression worked across tissues.

The tissue-specific results support generated expression as a regularizer or feature prior rather than independent biological sample size. Every biological association was tested in real profiles after estimating FLT/GC effects separately by accession. Synthetic-promoted and reinforced labels describe repeated feature selection and do not claim biological novelty.

In thymus, synthetic guidance organized real-supported genes into an interpretable cell-cycle panel. The strongest promoted genes and the reinforced *Ube2c* and *Gmnn* shared lower flight effects and converged on mitosis and DNA replication. This is a developmental biological hypothesis, not evidence that generated profiles add animals or independent statistical support.

Soleus produced reinforcement rather than discovery. The synthetic arm retained the same five BH-FDR genes as stable real-only selection and sharpened balanced accuracy without improving AUROC. Independently selected genes converged on mitochondrial turnover and fatty-acid metabolism; the model did not nominate a new soleus gene.

Kidney provides the clearest promotion result outside thymus. Synthetic guidance added *Inpp4b* to reinforced *Slc37a4*, and the real-data association for *Inpp4b* survived leave-one-accession-out analysis. Spleen provides a weaker multi-gene example: *Rai14*, *Ptprk*, and *Myl9* were promoted and *Loxl1* reinforced, but none passed LOO FDR and no pathway was significant. Quadriceps and EDL selected real-only arms and therefore do not support synthetic-guided claims.

Thymus and soleus define the clearest biological hypotheses

The thymus result refines established spaceflight immune biology. Prior work documents thymic involution and altered cell-cycle expression [8,9]. The current signature concentrates on cyclins, CDK1, UBE2C, BIRC5, NUSAP1, geminin, APC/C-mediated protein turnover, and G2/M control. Together these support lower proliferative renewal or a lower proportion of cycling thymocytes. Because the data are bulk, composition and cell-intrinsic regulation remain inseparable.

The soleus result addresses a different physiological axis. Weight-bearing slow muscle is especially sensitive to unloading. Lower *Bdh1*, *Ech1*, *Bnip3*, and *Decr1* together with higher *Tpm1* describe reduced oxidative substrate handling, altered mitochondrial quality control, and contractile remodeling. This is compatible with known soleus atrophy, altered lipid metabolism, and slow-to-fast transition [10,11]. Synthetic guidance reinforced this compact fatty-acid-oxidation panel across soleus studies.

Secondary and negative tissue results

The other tissues constrain the method's scope. Kidney combines one promoted and one reinforced gene but lacks pathway enrichment. Spleen combines strong within-study predictive gains with four BH-FDR genes, yet none survives the stricter LOO criterion. Skin promotes one directionally consistent gene while broader pathway enrichment remains heterogeneous. Adrenal gland has only three studies. Retina and lung have predictive or pathway gains without a synthetic-informed BH-FDR gene. Liver, quadriceps, and EDL selected real-only arms in development.

A new thymus study should test the ten-gene mitotic panel. Soleus experiments should quantify the five-gene mitochondrial and contractile panel together with fatty-acid oxidation. Kidney should test *Inpp4b* and *Slc37a4* jointly. Spleen should test *Rai14*, *Ptprk*, *Myl9*, and *Loxl1* in stroma-preserving assays.

Limitations

The tissue screens split profiles within represented accessions. They measure development utility but do not establish performance in a new mission, strain, or processing protocol. Agreement across represented real studies supports focused follow-up, and independent datasets are still required for confirmation.

The configurable matrix was executed as a gated search, not as a full factorial benchmark. Selected shared and paper-based preprocessing arms, nine liver harmonization arms, study-conditioning policies, and the leading WGAN and DDIM configurations were trained to completion. Consequently, the study identifies a validated model path rather than a universal optimum across every possible transformation and hyperparameter combination. WGAN-GP and DDIM final values also come from different selection stages because opening the WGAN locked test after failed validation would have violated the protocol.

The analysis used a 974-gene landmark panel, so relevant genes outside that panel could not be discovered. BH FDR was controlled separately within each declared tissue family, not once across every tissue-gene combination in the project. Direction disagreement and high heterogeneity do not remove a BH-significant association, but they reduce confidence that its pooled effect represents a common response across missions. Bulk tissue also cannot distinguish a transcriptional change within a cell type from a change in cell composition. The thymus result could reflect lower expression in proliferating thymocytes, fewer proliferating thymocytes, or both. The spleen structural candidates likewise cannot distinguish altered expression from altered stromal or contractile-cell abundance.

Finally, the generator represents tissue, condition, accession, and material contexts available during adaptation. The ARCHS4 reference failed its strict correlation-structure gate, while the adapted model passed the finite-sample fidelity rule on a locked within-study test. These results do not show that the model will reproduce an unrepresented mission, strain, or sample-processing protocol. Exact model limitations, sensitivity analyses, and statistical safeguards are documented in the supplementary methods.

Conclusions

Synthetic expression was useful only when its role was selected for the tissue and evaluated against real-only training. A pooled augmentation policy failed. Thymus and soleus provide the clearest process-level hypotheses, while kidney *Inpp4b* and *Slc37a4* remain focused secondary candidates.

Skin, spleen, pooled skeletal muscle, adrenal gland, gastrocnemius, and tibialis anterior provide narrower developmental hypotheses. Lung and retina improved during development but lacked a corresponding synthetic-informed BH-FDR gene in the landmark panel. Quadriceps, EDL, and liver do not support synthetic-guided claims because their real-only arms were retained. Independent biological data are needed to test all of these hypotheses.

Data and code availability

OSDR data were accessed through the public Biological Data API [1]. ARCHS4 and Reactome are public resources [2,7]. Repository scripts are under `src/nasa_mouse_rna_diffusion/`; frozen run outputs are under `outputs/generative_benchmark/`; this paper package is under `paper/synthetic_guided_spaceflight/`. The figure and table builder is `nasa_mouse_rna_diffusion.build_synthetic_guided_paper`. Public repository URL and archival DOI will be added after author review.

Ethics statement

This study is a secondary computational analysis of publicly available animal-study data. No new animals were used.

Competing interests

The author declares no competing interests.

Acknowledgments

The author acknowledges the NASA Open Science Data Repository, GeneLab data-processing teams, original flight-study investigators, ARCHS4, Reactome, and the developers of the evaluated generative-model implementations. Program and mentor acknowledgments require final author review.

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